

Coco Choi

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EDUCATION

Northeastern University, Boston MA / San Jose CA

Sep 2024 – May 2027

Master of Science in Computer Science

GPA: 3.8/4.0

- Related courses: Object Oriented Design, Data Structures and Algorithms, Web Development, Advanced Software Development, Database Management Systems, Cloud Computing, Computer Network, Distributed Systems, Computer Systems, Machine Learning

Franklin & Marshall College, Lancaster PA

Aug 2018 - Dec 2021

Bachelor of Arts in Business, Administration and Management / Film and Media Double Majors

GPA: 3.72/4.0

TECHNICAL KNOWLEDGE

- **Languages:** Python, Java, JavaScript, TypeScript, C++, SQL, NoSQL, Go (Golang), Rust, Kotlin, Swift, C#, R, Ruby, Linux/Unix
- **DevOps & Cloud:** Amazon Web Services (AWS EC2, S3, Lambda, SageMaker), Docker, Kubernetes, Jenkins, CI/CD
- **Backend:** Spring Boot, Node.js, Express, Django, Flask, ASP.NET Core, RESTful APIs, Microservices, GraphQL, gRPC, OOP
- **Database & Caching:** MySQL, PostgreSQL, MongoDB, Redis, Elasticsearch, Snowflake, DynamoDB, Cassandra, Memcached

WORK EXPERIENCE

Amazon, AWS Global Backbone, Software Development Engineer Intern, Santa Clara CA

Aug 2026 – Dec 2026

- Built a **Java** traffic-replay simulator ingesting **14 days BGP-LS/gNMI** snapshots across **1,200+ backbone links** on **EKS**; regression-tested **200+ congestion scenarios** against **TE controller**, surfacing **4 path-solver corner cases** and cutting validation cycle by **60%**.
- Implemented a weighted-**ECMP** heuristic in **C++** for the **TE controller path solver (Google OR-Tools)**, cutting link-utilization variance by **18%** across **8 regions** and **40+** peering edges; added **gRPC** hooks, passing **95%** lab validation on **500+ TE LSPs**.
- Engineered a **Java/Kafka** link-utilization aggregator ingesting **gNMI telemetry** from **2,500+** backbone interfaces at **5s** granularity; published via **gRPC**, cutting decision-loop latency from **12 s to 3 s** and lifting congestion-detection recall by **22%**.
- Designed a **Python/Kubernetes** failure-injection harness simulating **30+** link-cut/node-failure scenarios on a containerized **SDN-fabric emulator (EKS)**; validated **FRR** convergence under **200 ms** on **95%** of **400+** runs, catching **3 failover race conditions**.
- Shipped a what-if analysis dashboard (**React + FastAPI on EKS**) visualizing **TE path decisions** across **1,000+** backbone edges and **60+ POPs**; integrated **OpenTelemetry** traces and **8 metric views**, adopted by **5 NDEs** and cutting capacity-planning prep by **35%**.

Volkswagen, Audi Connected Vehicle & Digital Services, Data Engineer (Full-time), Beijing China

Jan 2023 - Aug 2024

- Ingested **500+ GB/day** from **20K+** IoT sensors via **Kafka** with **99.9%** uptime on **Spring Boot/Flask microservices**; implemented horizontal sharding and round-robin load balancing, increasing throughput **3x** and reducing ingestion failures by **75%**.
- Designed high-performance **ETL** with **Apache Airflow** and **Spark Structured Streaming**; parallelized **Pandas/NumPy** transforms, reaching sub-second latency and reducing batch processing time by **42% (6 h to 3.5 h)** across **12 teams**.
- Designed **Snowflake** warehouse with **dbt**, improving **SQL performance by 60%**; delivered **15+ PySpark** predictive-maintenance data science solutions associated with **25%** higher customer retention (**about \$2M/year**), methodology available upon request.
- Implemented infrastructure as code using **Docker**, **Kubernetes**, and **GitHub Actions** for **CI/CD**; achieved **90%** test coverage and collaborated with **PM, UX** and **8 engineers** using **Agile methodologies**, shipping **two sprints** ahead of schedule.

Mars Inc., Digital Commerce APAC, Cloud Engineer Intern, Hong Kong

Jun 2022 – Dec 2022

- Built **containerized** e-commerce backend with **Node.js/Express GraphQL**, **OAuth 2.0/JWT** authentication, and **RBAC** on **AWS ECS /Fargate**; served **5,000+ DAU** at **99.9% uptime** with **auto-scaling** architecture supporting **\$300K Q1 revenue**.
- Scaled messaging with **Socket.IO**, **Redis pub/sub**, **RabbitMQ**; implemented distributed rate limiting (token bucket and sliding window data structures) and event sourcing to handle **10K** concurrent connections during **2x baseline** peak traffic.
- Reduced search latency by **65%** in **MongoDB** via **Mongoose ODM** geospatial and compound indexes, plus **Redis** caching (**90%** hit rate); improved maintainability by **30%** with **OOP** design patterns and separation of concerns under **Agile, Git**.

WPP plc, GroupM Nexus, Cloud Data Platform Intern, Hong Kong

Jan 2022 – May 2022

- Engineered **Python/Go** server-side ad-event ingestion via **Meta CAPI**, **Google Enhanced Conversions**, and **TikTok Events API** across **25+ platforms**; streamed **500M+ events/day** through **AWS Kinesis Firehose/Glue** into **Parquet S3** at **99.95% delivery**.
- Built identity-resolution graph unifying **80M+** profiles across **HEM/MAID/IDFA/CRM** via deterministic + probabilistic matching (**Splink/LSH**) at **92% OneID**; enforced **GDPR/CCPA/PIPL** via **IAB TCF 2.0**, k-anonymity, differential privacy, **AWS Lake Formation**.
- Architected **AWS Redshift** and **Spectrum** with **Kimball star schema**, **SCD Type 2**, and **120+ dbt models** orchestrated via **AWS Step Functions/Athena**; enforced data quality via **Great Expectations**, cut BI latency **75% (60s to 15s)**, and storage cost **40%**.

RESEARCH EXPERIENCE

Northeastern UbiWell Lab, Graduate Research Assistant, Boston MA

May 2025 - Oct 2025

- Designed and built scalable **Python (Django/FastAPI)** microservices with **PostgreSQL/MongoDB (NoSQL)**, processing **2.5M+** data points/day with **p95 under 200 ms** and **99.9%** availability; exposed **REST (JSON)** and **gRPC (Protocol Buffers)** APIs.
- Optimized report generation time by **85% (8 h to 1.2 h)** through multi-tier caching and algorithm development; delivered **100+ React/TypeScript** dashboards (**D3.js, Plotly**) with real-time, role-based views for **1,000+** daily research users.
- Built fault-tolerant scalable pipelines collecting **500+ hours** of patient-monitoring data via APIs and **BeautifulSoup/Selenium**; integrated **Apache Kafka** event-driven architecture, **peaking at 50K events/s**, and reduced data loss by **15%**.

PROJECTS

Personalized News Summarization RAG Platform, Northeastern University

Oct 2024 - April 2025

- Built real-time news summarization **RAG** pipeline over **50K+ articles**; combined **BM25** with **bge-large** embeddings in **pgvector** on **AWS RDS** plus **Cohere Rerank v3**, lifting retrieval **nDCG@10** by **35%** and answer **faithfulness** to **92%** on **2K+ eval queries**.
- Fine-tuned **bge-large-en** embeddings on **120K** pairs via **contrastive learning (PyTorch + DeepSpeed)** on **AWS SageMaker (4 × A10G GPUs)**, cutting training **67% (12 h to 4 h)** and lifting accuracy **15%**; enhanced queries with **HyDE** and **spaCy NER**.
- Built **LLM** observability with **Prometheus/Grafana**, **OpenTelemetry**, **LangSmith** traces, and **ELK** stack at **99.95% SLA (30-day rolling)** while optimizing token usage and **vLLM** inference batching, cutting compute **\$8K/month** and on-call escalations **60%**.